# Ming Gong, Ph.D.

Bellevue, WA | +1 206 833 0010 | yiming10.13@163.com | Google Scholar

## Summary

Seasoned **applied-sciences and full-stack engineering leader** with a proven track record of building consumer-scale **search, ranking, recommendation, and Generative AI** products. Currently Head of ML Engineering at Atlassian, serving as the single point of accountability for an extended org of **130+** across Jira Search & Agent, Knowledge Fundamentals, and Generative Answers & AI Search Experiences — leading Jira Search's transformation and **Generative AI Search from zero to GA**. Previously Group Senior Principal Applied Sciences Manager at Microsoft, built and led a 40+ team that took **Bing Search from traditional ML through deep learning to Agentic AI**: built Bing Question-Answering from zero, scaled it to **100+ languages and 230+ markets**, shipped Knowledge Card (**65M DAU**), entity recommendation systems that **beat Google on relevance**, and **Copilot Search as a Bing AI vertical one month ahead of Google's AI Mode**.

End-to-end search and AI expertise spanning query understanding, search relevance & ranking modeling, recommendation systems & signals, extractive and generative answers, agentic search frameworks & experiences, personalization, LLM fine-tuning & post-training, and large-scale ML serving. Uniquely multi-dimensional: **applied sciences & full-stack engineering**, **AI products & research** (50+ publications, 11K+ citations, 20+ patents), **consumer & enterprise products**.

## Selected Highlights

- **Search ranking evolution — ML → DL → Agentic AI:** At Microsoft, built Bing QnA from the ground up and led its evolution through **every major search paradigm shift** — from GBDT rankers (400+ hand-crafted features) to DNN/BERT-based ranking models (10+ releases that won WPSBS vs. Google) to Generative SERP and **Copilot Search** (agentic, multi-turn AI search vertical). At Atlassian, leading Jira Search transformation and **Generative AI Search experiences from zero to GA**.
- **Search & QnA system from 0 → global (100+ languages):** Built Bing's Question-Answering system from scratch — a core search ranking product — and scaled it from English-only to **100+ languages and 230+ markets** via systematic cross-lingual methodology. Placed the universal multilingual bet ∼**1 year before globalization became an official org strategy**. Universal model served 100+ languages on ∼**13% of per-language GPU footprint**. **2× Microsoft MLADS Distinguished Contribution Awards** (first-ever back-to-back; CTO Office global awards).
- **Ranking & recommendation at consumer scale:** Knowledge Card on Bing — 150M entities, 60% query traffic, **65M DAU**, +3.0% WPSBS; certified as a Bing "WoW" feature (<10 out of 100+ features). Segment Entity Recommendation (recipe, job, movie) — **beat Google on relevance**. Jira Search — **3.4× Search MAU growth (2.5M → 8.5M)**.
- **Generative AI:** Led Copilot Search (multi-turn AI search vertical) — **shipped one month before Google's AI Mode**. Built Glow SLM framework replacing GPT-4-class models (**20–30 s → ∼5 ms latency**). Generative Answers covering **20% of Bing global queries**, +2.0% WPSBS. At Atlassian, Generative Answers and Enterprise GenAI Search drove **AI MAU from 100K → 1.2M**.
- **Ship velocity:** Jira Search MVP shipped **4 months** after joining Atlassian, full version launch within 12 months; Enterprise GenAI Search POC in **<1 month**, MVP GA in early 2026. First BERT model deployed in Bing **within 1 month of open-source**.
- **Org scale:** Single POC for an **extended org of 130+** at Atlassian (60+ direct + dotted-line partner teams); previously built a 40+ team at Microsoft from scratch across two sites.
- **Research recognition:** Best Paper Award, IJCAI '24 AI4Research; 50+ publications, 11K+ citations, 20+ patents.

## Experience

**Atlassian (US)** — **Feb 2025 – Present**
*Head of Machine Learning Engineering — Knowledge AI* — *Bellevue, WA*

- Single point of accountability for the Knowledge AI charter, leading an **extended org of 130+** — 60+ direct reports (70% scientists, 30% engineers; full-stack and globally distributed) plus dotted-line partner teams across Jira Search & Agent, Knowledge Fundamentals, and Generative AI Search. Grew the team from 30 to 60+ in one year, recruiting **2 Principal-level Managers and 3 Principal-level Tech Leads**; established new teams in **Canada and Singapore** (greenfield sites for the company).
- **Search ranking & recommendation:** Shipped Jira Search MVP within 4 months of joining, full version launch to all customers within 12 months; rebuilt ranking stack and search UX, driving **3.4× Search MAU growth (2.5M → 8.5M)**, helping AI Org exceed Search MAU L1 OKR; premiered at Team '25. Formed a **V-Team growing from ∼30 to 115 contributors across 8 partner orgs** with a Coalition Council to drive cross-org alignment.
- **Knowledge signals & entity recommendation:** Built core knowledge fundamentals — collaboration graph, entity linking, entity recommendation — now adopted by multiple product teams for personalized features. Led **entity linking turnaround from 40% to 90%+ precision**; invested 2 months in hands-on diagnosis, protected team stability through parallel senior hiring, then redirected technical approach.
- **Generative AI products:** Generative Answers in search and AI Definitions drove **AI MAU from 100K to 1.2M** via a novel generative UI framework. Initiated and led **Enterprise Generative AI Search** from zero to GA — identified a cross-product platform gap no team was chartered to own; POC in <1 month, MVP shipped in early 2026. The novel **Generative UI layer (Lumina) — 20% engagement uplift, 30% SBS lift** vs. competitor AI experiences — was adopted as Atlassian's standard generative presentation framework, now being integrated across all GenAI products and features. Previewed at **Team '25 EU AI Super Session**; live demo at **Team '26 Founder Keynote**.

- **Agentic AI:** Launched **Jira Agents in Rovo Chat** within 3 months, becoming a primary driver of Rovo adoption. Code-driven agent architecture **beat Manus and GenSpark on the GAIA benchmark**.

**Microsoft — Search Technology Center Asia (Bing, Edge, Copilot)** **Aug 2013 – Dec 2024**
*Group Senior Principal Applied Sciences Manager (2019–2024) · Senior Dev Manager (2018–2019) · Senior Tech Lead (2016–2018) · SWE II (2013–2015)* *Beijing / Suzhou*

- **Built and led an org of 40+ engineers and applied scientists from scratch** (70% scientists, 30% engineers; **5 managers** — 3 science, 2 engineering) across two sites (Beijing and Suzhou), spanning Bing Search, Copilot, Cross-Lingual QA, Knowledge Graph, Recommendation, and Edge browser AI.
- **Bing Question-Answering: Zero-to-Global, Traditional ML $\rightarrow$ DL $\rightarrow$ Pre-trained Models $\rightarrow$ LLM (2016–2024):** built Bing QnA — a core search ranking and relevance product — from the ground up and led its evolution through every major AI paradigm shift. Placed the universal multilingual bet $\sim$**1 year before globalization became an official org strategy**. Scaled from English-only to **100+ languages and 230+ markets** via systematic cross-lingual methodology — **2$\times$ MLADS Distinguished Contribution Award** (first-ever back-to-back recipient; CTO Office global awards). Shipped 10+ DNN releases that replaced legacy GBDT ranker (400+ hand-crafted features) and won WPSBS vs. Google. Co-owned NRT inference pipeline design with **8 platform teams across time zones**, treating partners as co-owners with shared credit — pipeline became Platform team's flagship demo and blueprint for subsequent products. Authored novel model compression that powered the **first BERT model deployed in Bing** (within 1 month of BERT open-source, zero additional GPU cost) — +3.0% coverage, +5.0% precision. Universal model served 100+ languages on $\sim$**13% of the GPU footprint** a per-language approach would have required.
- **Copilot Search — Agentic AI search vertical (2024–2025):** delivered multi-turn AI search experience unifying GenQnA, GenSERP, and Magazine; built scalable LLM generation pipelines and task-specific SLMs (incl. R1-distilled). **Shipped April 2025 — one month before Google's AI Mode.**
- **Generative SERP (2023–2024):** reinvented Bing SERP as "AI Masterpieces" — proposed tier-wise hallucination control framework **against split organizational consensus** (conservative vs. velocity camps); held the position through 5–6 leadership reviews over two months. Framework adopted by Edge Copilot and Office Copilot. Built **Glow**, a distilled SLM framework replacing GPT-4-class models — first-token latency **20–30 s** $\rightarrow$ $\sim$**5 ms**; V2 scaled to 5–10% query coverage.
- **Generative Answers on Bing SERP (2023–2024): 20% query coverage on Bing global markets, +2.0% WPSBS** via streaming generative answers and click-through to Bing Chat.
- **Knowledge Card — ranking & recommendation for 150M entities (2021–2023):** multi-modal entity aggregation across web sources with template + MRC DL extraction and LLM generation. **60% Bing query traffic, 65M DAU, +3.0% WPSBS**; certified as a Bing "WoW" feature (<10 out of 100+ features).
- **Segment Entity Search & Recommendation (2021–2024):** shipped recipe, job, and movie recommendation answers — **beat Google on relevance and WPSBS** on recipe and job; closed WPSBS gap with Google on movie recommendation. End-to-end ranking pipeline from content signals to relevance modeling and delivery.
- **Query-Dependent Passage Extraction (2019–2022):** replaced industry-standard sliding-window passage indexing with runtime query-dependent extraction via GNN document modeling + multi-teacher distillation to 40ms latency. **Held #1 on Google Natural Questions benchmark for** $\sim$**1 year**; architecture adopted across Bing QnA, Knowledge, People Also Ask, and Azure Custom QnA.
- **Microsoft Copilot on Edge (2023–2024):** long-document summarization and chat over PDFs and long-form web pages — 1000+ page docs at **95%+ accuracy**; demoed at Microsoft Copilot 2023 announcement event; shipped to Edge and Windows Copilot.
- **Underside on Edge — Right-Rail Insight Pane (2022–2024):** built the AI-powered right-rail panel for Edge featuring page summary, topics, QnA, related search, and related pages (feeds). **Edge engaged DAU +10%, Search DAU +1.2M.**
- **NLU Platform "Orca" (2020–2025):** unified query understanding pipeline (domain, intent, slots, topics) powering answer triggering. **95%+ of Bing answers (coverage $\geq$ 0.05%)** onboarded; 60% of onboarding answers closed triggering gap with Google.

## Platforms & Internal Tooling Built

- **Lumina** (Atlassian, 2025–2026) — generative UI framework for Enterprise GenAI Search; **20% engagement uplift, 30% SBS lift**; adopted as Atlassian's standard generative presentation framework across all GenAI products.
- **X-SuperAgent** (Atlassian, 2025–2026) — Atlassian's first OpenClaw-compliant agent framework for AI-native team operation; **40%+** developer-productivity uplift.
- **Glow** (Microsoft, 2023–2024) — distilled SLM serving framework replacing GPT-4-class models for production; first-token latency **20–30 s** $\rightarrow$ $\sim$**5 ms**; V2 scaled to 5–10% Bing query coverage.
- **Orca** (Microsoft, 2020–2025) — unified NLU platform (domain, intent, slots, topics) powering answer triggering for **95%+ of Bing answers**.
- **NeuronBlocks** (Microsoft, 2019, open source) — "Build NLP DNN models like playing Lego." Published at EMNLP/IJCNLP 2019.

## People & Org Development

- **Founder & Head, Microsoft AI School — Asia-Pacific R&D Group** (2017–2024). Led a **50+ operation v-team**; drove planning, curriculum, and execution of all AI learning events for the region. **10K+ participants, 2K+ honored graduates**, many of whom grew into AI talents and leaders across the Microsoft APRD org (with strict graduation criteria); record-breaking participation 8 years running.
- **Invited instructor, LinkedIn Global Academy for Chinese Programmers** (2022–2023): "Intelligent QA Systems for Search Engines" — **50M impressions, 98% satisfaction**.

- **Invited speaker, InfoQ AICon Global AI Conference** (2021): represented Microsoft China on globalization of Bing QA.

## Education

**Ph.D., Graphics and Visual Computing** 2008 – 2013
Institute of Computing Technology, Chinese Academy of Sciences

## Selected Awards

- **Best Paper Award**, IJCAI '24 AI4Research — *"Step-Back Profiling: Distilling User History for Personalized Scientific Writing."*
- **2021 Fall Microsoft MLADS Distinguished Contribution Award** (2 / 250+). Broke conference history to receive the award twice consecutively.
- **2021 Spring Microsoft MLADS Distinguished Contribution Award** (2 / 300+).
- 2023 Microsoft Global Hackathon — Edge Theatre: **Best Overall Hack**; Greater China Region: **Most Popular Hack**.
- 2023 Microsoft Global Hackathon — DeKG (decentralized knowledge ecosystem): **Best Innovativeness & Originality Hack**; Beijing Outstanding Project Award.
- Excellent Operation Award of AI School China — 2018, 2019, 2020, 2021, 2022, 2023.
- Microsoft AI Core Q3 FY18 **Greatness Award** (2018).

## Selected Publications (50+ papers, 11,000+ citations on Google Scholar — full list on scholar profile)

### ▷ Recommendation & Personalization

- Ning Wu, **Ming Gong**, Linjun Shou, Jian Pei, Daxin Jiang. "RUEL: Retrieval-Augmented User Representation with Edge Browser Logs for Sequential Recommendation." *CIKM* 2023.
- Hanshuang Tong, Jun Li, Ning Wu, **Ming Gong**, et al. "Ploutos: Towards Interpretable Stock Movement Prediction with Financial Large Language Model." *arXiv* 2024.

### ▷ Dense Retrieval & RAG

- Shengyao Zhuang, Linjun Shou, Jian Pei, **Ming Gong**, et al. "Typos-aware Bottlenecked Pre-Training for Robust Dense Retrieval." *SIGIR* 2023.
- Houxing Ren, Linjun Shou, Jian Pei, Ning Wu, **Ming Gong**, Daxin Jiang. "Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval." *EMNLP* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Modeling Sequential Sentence Relation to Improve Cross-lingual Dense Retrieval." *ICLR* 2023.
- Shunyu Zhang, Yaobo Liang, **Ming Gong**, Daxin Jiang, Nan Duan. "Multi-View Document Representation Learning for Open-Domain Dense Retrieval." *ACL* 2022.
- Linjun Shou, S. Bo, Fei-Xiang Cheng, **Ming Gong**, et al. "Mining Implicit Relevance Feedback from User Behavior for Web Question Answering." *KDD* 2020.

### ▷ LLM, Agentic AI & Multimodal Pre-training

- Yaobo Liang, Chenfei Wu, Ting Song, et al., **Ming Gong**, Nan Duan. "TaskMatrix.AI: Completing Tasks by Connecting Foundation Models with Millions of APIs." *Intelligent Computing* 2023.
- Gen Li, Nan Duan, Yuejian Fang, **Ming Gong**, Daxin Jiang. "Unicoder-VL: A Universal Encoder for Vision and Language by Cross-modal Pre-training." *AAAI* 2019.
- Nuo Chen, Ning Wu, Shining Liang, **Ming Gong**, et al. "Is Bigger and Deeper Always Better? Probing LLaMA Across Scales and Layers." *arXiv* 2023.
- Ning Wu, **Ming Gong**, Linjun Shou, et al. "Large Language Models are Diverse Role-Players for Summarization Evaluation." *NLPCC* 2023.

### ▷ Model Training, Compression & Distillation

- **Ming Gong**, Linjun Shou, Wutao Lin, et al. "NeuronBlocks — Building Your NLP DNN Models Like Playing Lego." *EMNLP/IJCNLP* 2019.
- Ze Yang, Linjun Shou, **Ming Gong**, Wutao Lin, Daxin Jiang. "Model Compression with Two-stage Multi-teacher Knowledge Distillation for Web Question Answering System." *WSDM* 2020.
- Fei Yuan, Linjun Shou, Jian Pei, Wutao Lin, **Ming Gong**, Daxin Jiang. "Reinforced Multi-Teacher Selection for Knowledge Distillation." *AAAI* 2021.

### ▷ Code Intelligence

- Zhangyin Feng, et al., **Ming Gong**, Linjun Shou, et al. "CodeBERT: A Pre-Trained Model for Programming and Natural Languages." *EMNLP (Findings)* 2020. (3,000+ citations)
- Junjie Huang, Duyu Tang, Linjun Shou, **Ming Gong**, et al. "CoSQA: 20,000+ Web Queries for Code Search and Question Answering." *ACL* 2021.

### ▷ Cross-Lingual & Multilingual Modeling

- H. Huang, Yaobo Liang, N. Duan, **Ming Gong**, et al. "Unicoder: A Universal Language Encoder by Pre-training with Multiple Cross-lingual Tasks." *EMNLP/IJCNLP* 2019.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei, Daxin Jiang. "Bridging the Gap between Language Models and Cross-Lingual Sequence Labeling." *NAACL* 2022.
- Nuo Chen, Linjun Shou, **Ming Gong**, Jian Pei. "From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension." *AAAI* 2022.

- Shining Liang, **Ming Gong**, Jian Pei, Linjun Shou, Daxin Jiang. "Reinforced Iterative Knowledge Distillation for Cross-Lingual Named Entity Recognition." *KDD* 2021.
- Nuo Chen, Linjun Shou, Tengtao Song, **Ming Gong**, et al. "Structural Contrastive Pretraining for Cross-Lingual Comprehension." *ACL* 2023.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Language Scaling: Applications, Challenges and Approaches." *KDD Tutorial* 2021.
- Linjun Shou, **Ming Gong**, Jian Pei, et al. "Scaling out NLP Applications to 100+ Languages." *WWW Tutorial* 2021.

▷ **Knowledge & Entity Understanding**

- Zilin Xiao, **Ming Gong**, Jie Wu, et al. "Instructed Language Models with Retrievers Are Powerful Entity Linkers." *EMNLP* 2023.
- Zilin Xiao, Linjun Shou, Xingyao Zhang, Jie Wu, **Ming Gong**, et al. "Coherent Entity Disambiguation via Modeling Topic and Categorical Dependency." *Findings of EMNLP* 2023.

## Selected Patents (20+ granted; selection below)

- Personalized Multilingual Content Recommendation based on Robust Feature Network. 2023. (MS#412659-CN-NP)
- Sequential Recommendation based on Cross-Domain Behavior Data. 2023.
- Sequential Recommendation Based on Dual Contrastive Interest Learning. 2022.
- Content Recommendation Based on Graph Enhanced Collaborative Filtering. 2022.
- Knowledge-enhanced Content-Aware Collaborative Filtering for Recommendation. 2021.
- Hierarchical Representation Learning of User Interest. 2021. (MS#410462-CN-NP)
- Model Generation based Model Compression. 2019. (MS#406805-CN-NP)
- Model Selection Learning for Knowledge Distillation. 2020. (MS#408426-CN-NP)
- Multi-Model Joint Denoising Training. 2021. (MS#409566-CN-NP)
- Contrastive Learning for Question Answering. 2020. (MS#407988-CN-NP)
- Providing QA Training Data and Training a QA Model based on Implicit Relevance Feedbacks. 2020. (MS#407927-CN-NP)
- Question Generation from Queries. 2021. (MS#410241-CN-NP)
- Assertion-based Question Answering with Question-Aware Open Information Extraction. 2019.
- Representation Learning of Semi-structured Data. 2020. (MS#408908-CN-NP)
- Graph based Fact Clustering for Diversity and Freshness. 2021.
- Lexicon-Enhanced Self-Supervised Training for Multilingual Dense Retrieval. 2022. (MS#412496-CN-NP)
- Sentence Representation Generation for Cross-lingual Retrieval. 2022.
- Representation Learning of Cross-Language Texts. 2021. (MS#409565-CN-NP)
- Cross-Lingual Task Training. 2019. (MS#406497-CN-NP)
- From Good to Best: Two-Stage Training for Cross-lingual Machine Reading Comprehension. 2021.
- Temporal Co-Contrastive Learning-Based Node Representation Generation. 2022.